

Mechatronics Programming for Real-time Systems (MCTR 1015)

Comprehensive Mechatronic Prosthetic Hand

Team 5

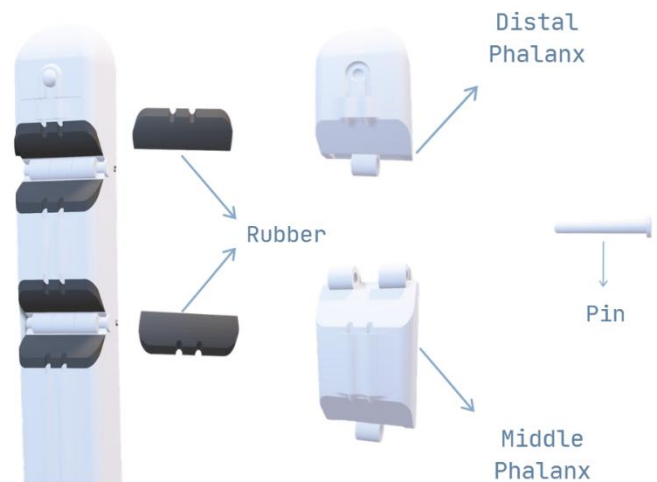
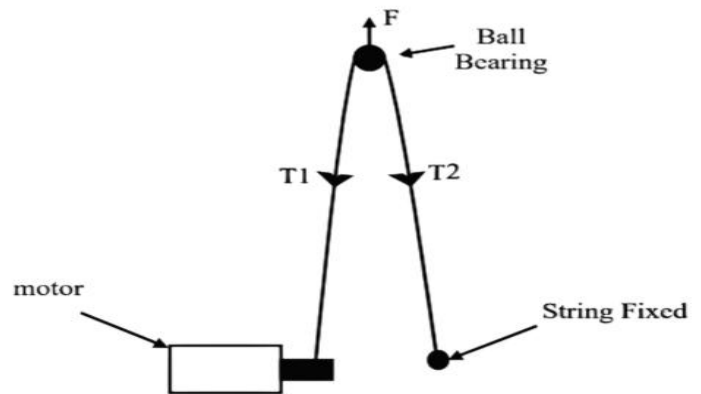
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1. Problem statement and Project Description

The prosthetic hand is a multi-domain electromechanical system designed to reproduce the flexion motion of human fingers using motor-driven tendon mechanisms. The system is powered by a 7.4 V LiPo battery, which supplies the DC motors through motor drivers and provides regulated voltage for the microcontroller and servo. The microcontroller receives the control signal and commands the motors and thumb servo to generate the required finger motions.

Each finger is actuated using a tendon-driven mechanism. A DC motor rotates through a gearbox to reduce speed and increase output torque. The motor shaft is connected to a spool/string system, where a single string pulls the phalanges of the finger. The string is routed over a pulley located at the distal phalanx and then fixed at the palm. This routing allows the force transmission to be distributed through the finger mechanism, while also reducing the effective torque requirement. The MCP, PIP, and DIP joints are modeled as pin-connected rotational joints between the phalanges. Rubber damping elements are placed between the phalanges to represent cartilage-like resistance and energy dissipation. In addition, elastic bands on the back of the fingers provide passive return motion and resist finger flexion, similar to biological tendons and ligaments.



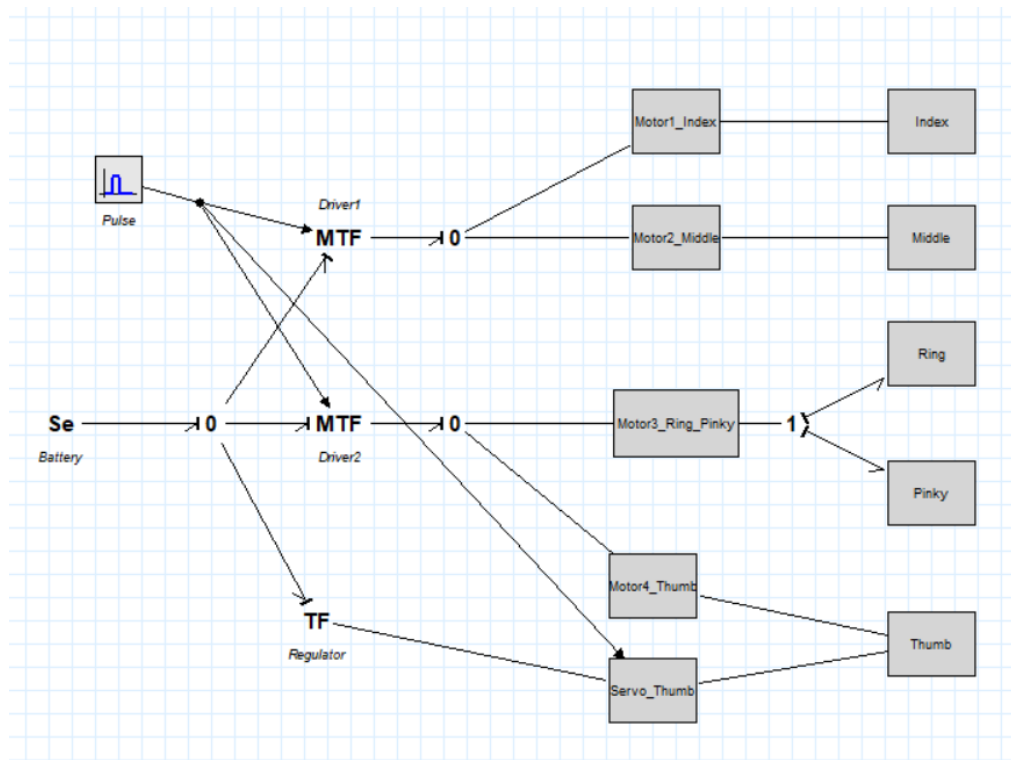
The actuation layout consists of one motor for the index finger, one motor for the middle finger, and one shared motor for the ring and pinky fingers. The thumb is modeled differently, since it requires two degrees of freedom. It uses a motor for tendon-driven motion and a servo motor to provide additional thumb motion. Unlike the other fingers, the thumb is not modeled with separate phalanx joints.

Because the prosthetic hand contains electrical, rotational mechanical, translational tendon, joint, damping, elastic, and gear transmission elements, deriving the complete mathematical model manually is complex and prone to errors. The system has many interacting components, multiple energy domains, and several energy-storing and dissipative elements. Therefore, Bond Graph modeling is used to represent the power flow between the electrical and mechanical domains in a structured way. This allows the system to be decomposed into submodels such as the battery, motor drivers, DC motors, gearboxes, tendon-string mechanisms, pulleys, joints, damping elements, elastic return elements, and thumb servo mechanism.

The objective of this project is to construct a complete Bond Graph model of the prosthetic hand and use it to derive the mathematical state-space representation of the system. This model will help describe how input voltage and control signals are converted into motor torque, tendon force, joint motion, and finger flexion. The resulting state-space equations can then be used to analyze the dynamic behavior of the prosthetic hand

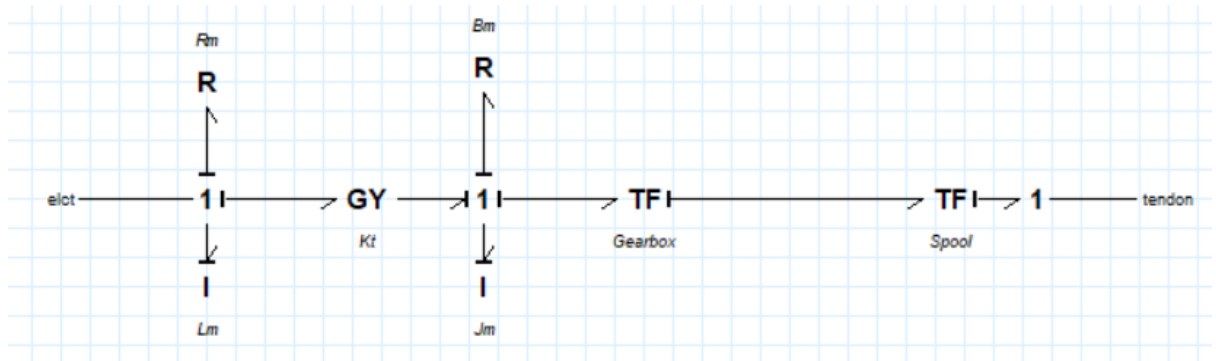
2. Flow Chart

The 20-Sim simulation of the model and showing all subsystems can be considered a simple flow chart of all connections



3. 20-Sim Model

- Motor model (For motors 1,2,3 & 4)



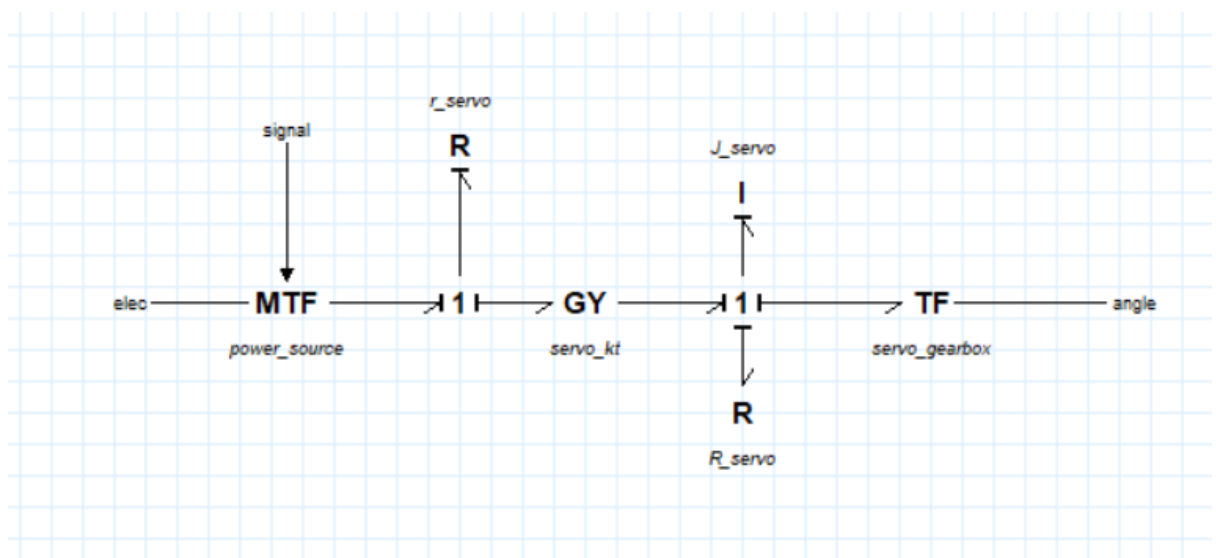
R_m : motor winding resistance - heat energy dissipated

L_m : motor inductance - magnetic energy stored

J_m : rotor inertia - rotational kinetic energy stored

B_m : motor damping/friction - mechanical energy dissipated

- Servo model

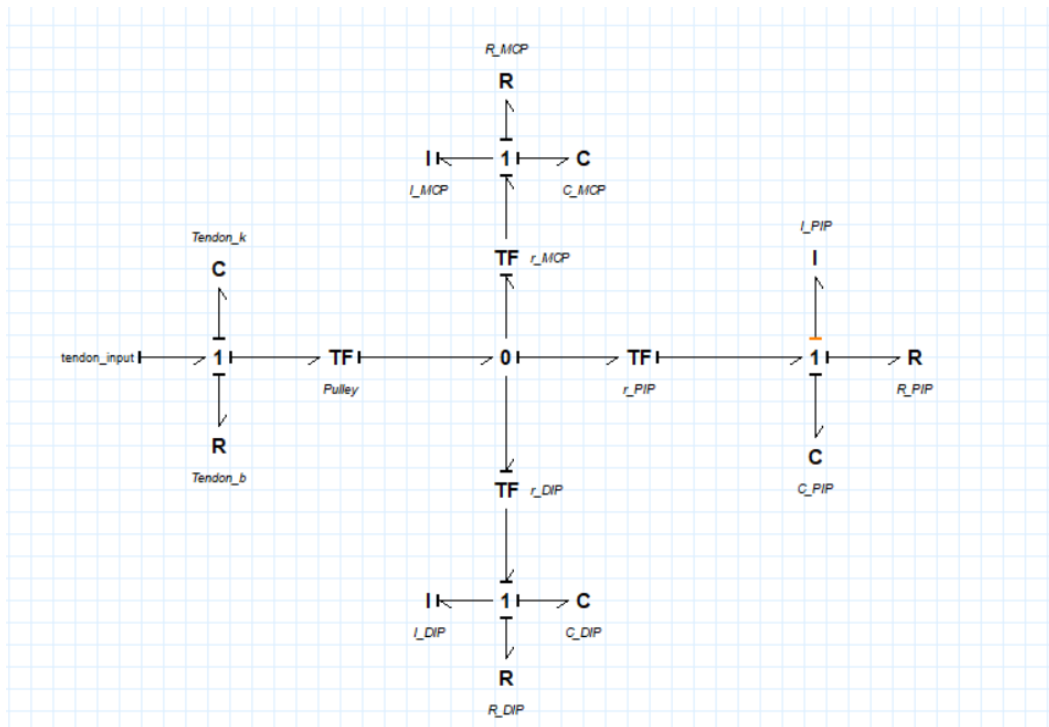


J_{servo} : servo shaft inertia - rotational kinetic energy stored

R_{servo} : servo damping/friction - mechanical energy dissipated

r_{servo} : servo internal resistance/damping - energy dissipated inside servo path

- Index, Middle, Ring & Pinky Fingers



Tendon_k: tendon stiffness - elastic energy stored in the tendon

Tendon b : tendon damping - mechanical energy dissipated due to tendon friction

I_MCP: MCP joint inertia - rotational kinetic energy stored

$\bar{C}$ MCP: MCP joint compliance/stiffness - elastic return energy stored

R_MCP: MCP joint damping - mechanical energy dissipated by rubber/cartilage

I_PIP: PIP joint inertia - rotational kinetic energy stored

$\bar{C}_{PIP}$: PIP joint compliance/stiffness - elastic return energy stored

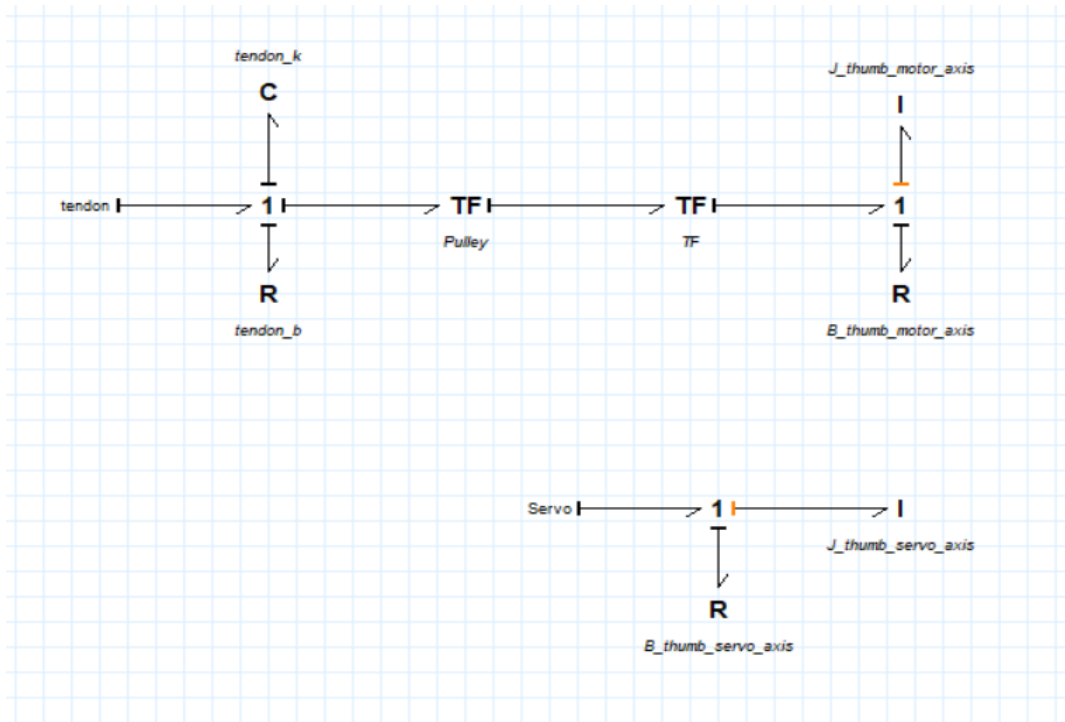
R_PIP: PIP joint damping - mechanical energy dissipated by rubber/cartilage

I_DIP: DIP joint inertia - rotational kinetic energy stored

$\bar{C}$ DIP: DIP joint compliance/stiffness - elastic return energy stored

R_DIP: DIP joint damping - mechanical energy dissipated by rubber/cartilage

- Thumb



tendon_k: thumb tendon stiffness - elastic energy stored in the tendon
tendon_b: thumb tendon damping - mechanical energy dissipated due to tendon friction
J_thumb_motor_axis: thumb motor-axis inertia - rotational kinetic energy stored
B_thumb_motor_axis: thumb motor-axis damping/friction - mechanical energy dissipated

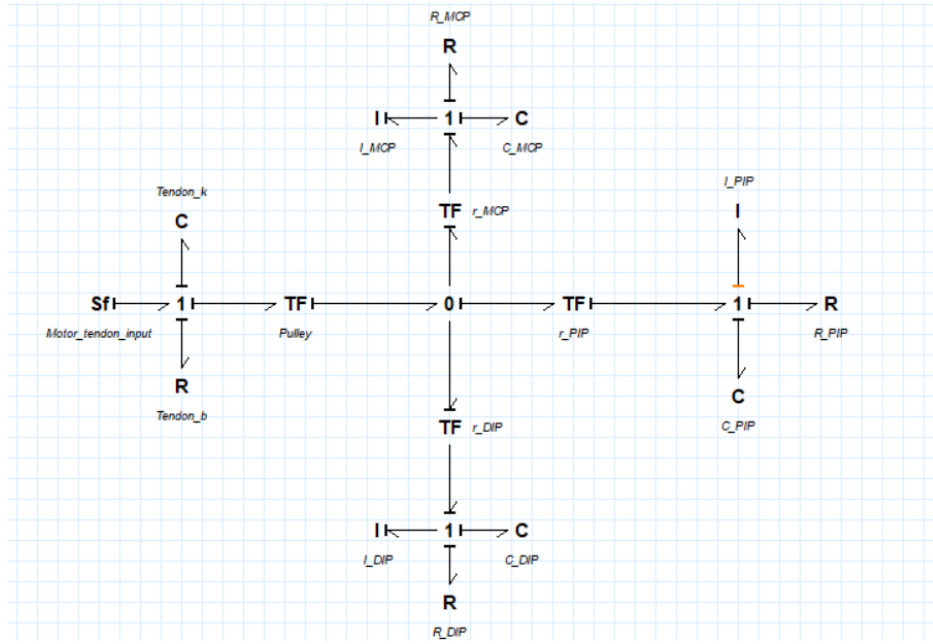
J_thumb_servo_axis: thumb servo-axis inertia - rotational kinetic energy stored
B_thumb_servo_axis: thumb servo-axis damping/friction - mechanical energy dissipated

4. Model Equations

As the whole prosthetic system was modelled not just the fingers the list of equations governing the model were very large in number. As well as the number of independent states

The model hand.emx contains:
117 submodels
190 equations
156 variables
34 independent states
The model has 0 errors and 9 warnings.

For simplification, only the equations for one submodel of one finger will be represented



The system equations given were

$$Tendon_k\backslash state = int(Tendon_k\backslash p.f, Tendon_k\backslash state_initial);$$

$$I_MCP\backslash state = int(I_MCP\backslash p.e, I_MCP\backslash state_initial);$$

$$C_MCP\backslash state = int(C_MCP\backslash p.f, C_MCP\backslash state_initial);$$

$$I_DIP\backslash state = int(I_DIP\backslash p.e, I_DIP\backslash state_initial);$$

$$C_DIP\backslash state = int(C_DIP\backslash p.f, C_DIP\backslash state_initial);$$

$$C_PIP\backslash state = int(C_PIP\backslash p.f, C_PIP\backslash state_initial);$$

State variable:

$$x_1 = Tendon_k\backslash state$$

$$x_2 = I_MCP\backslash state$$

$$x_3 = C_MCP\backslash state$$

$$x_4 = I_DIP\backslash state$$

$$x_5 = C_DIP\backslash state$$

$$x_6 = C_PIP\backslash state$$

State Space Equations:

$$\dot{x}_1 = u$$

$$\dot{x}_2 = -x_2 - \frac{2}{3}x_3 + \frac{1}{3}x_5 + \frac{1}{3}x_6 + \frac{1}{3}u$$

$$\dot{x}_3 = x_2$$

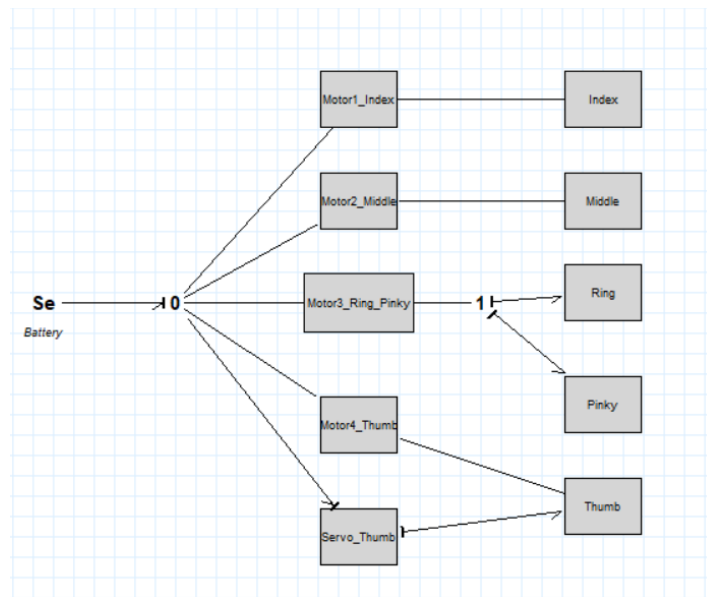
$$\begin{aligned}\dot{x}_4 &= -x_4 + \frac{1}{3}x_3 - \frac{2}{3}x_5 + \frac{1}{3}x_6 + \frac{1}{3}u \\ \dot{x}_5 &= x_4 \\ \dot{x}_6 &= -x_2 - x_4 + u\end{aligned}$$

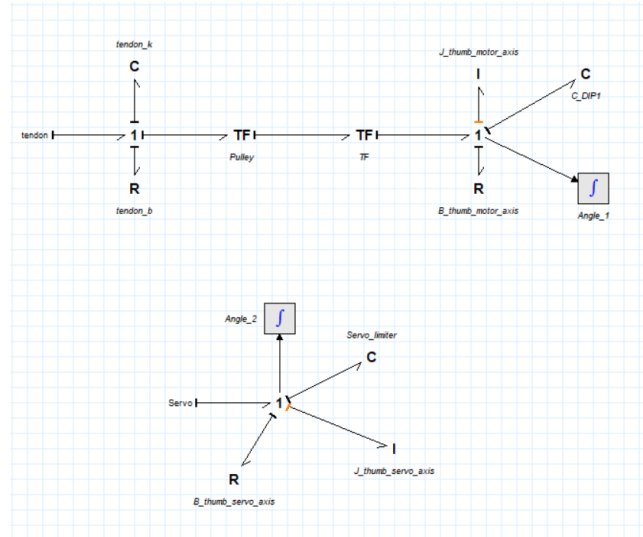
5. Proposed Validation Method

The proposed validation method is to compare the simulated joint angle response of the Bond Graph model with the measured response of the real prosthetic finger. A known motor input, such as a step PWM signal, is applied to the physical finger and the same input is applied to the 20-sim model. The MCP, PIP, and DIP joint angles are measured experimentally using video analysis or angle sensors, then compared with the simulated outputs C_MCP\state, C_PIP\state, and C_DIP\state. The accuracy of the model can be evaluated using steady-state error, rise time and settling time between the simulated and measured joint angles. If the error is large, the tendon stiffness, damping, joint inertia, and elastic band parameters can be tuned until the simulation response matches the experimental behaviour.

6. Milestone 3 modifications

The model was edited again and tested in simulation with parameters. For simplification only one finger was tested first and when it proved successful the whole hand was tested but the drivers were removed and the servo model was simplified to match a motor and the thumb sub-model was also improved.





7. System Parameters

The same motor parameter set is used for Motor1_Index, Motor2_Middle, Motor3_Ring_Pinky, Motor4_Thumb, and Servo_Thumb. The same finger parameter set is used for the index, middle, ring, and pinky fingers.

7.1 Input Battery Cases

Case	Battery input voltage (V)	Purpose
Case 1	12.0	Fully charged / high input case
Case 2	7.2	Nominal or partially discharged case
Case 3	3.7	Low battery case

7.2 Common Motor and Transmission Parameters

Parameter	Value	Description
Rm	4.5	Motor winding resistance / electrical damping
Lm	0.001	Motor winding inductance
Kt	0.00193	Motor torque constant / gyrator coefficient
Jm	5e-7	Rotor inertia
Bm	4e-6	Motor viscous friction
Gearbox r	0.012	Gearbox transformer ratio
Spool r	0.01	Spool transformer radius / ratio

7.3 Common Non-Thumb Finger Parameters

Parameter	Value	Description
Tendon_k	0.002	Tendon compliance/stiffness parameter used in the C element
Tendon_b	0.05	Tendon damping/friction
Pulley r	1	Pulley transformer ratio
I_MCP	0.00035	MCP joint inertia
I_PIP	0.00026	PIP joint inertia
I_DIP	0.00019	DIP joint inertia

C_MCP	11	MCP joint compliance/stiffness element
C_PIP	13	PIP joint compliance/stiffness element
C_DIP	14	DIP joint compliance/stiffness element
R_MCP	0.048	MCP joint damping/friction
R_PIP	0.036	PIP joint damping/friction
R_DIP	0.027	DIP joint damping/friction
r_MCP	125	MCP tendon moment-arm / transformer factor
r_PIP	142	PIP tendon moment-arm / transformer factor
r_DIP	166	DIP tendon moment-arm / transformer factor

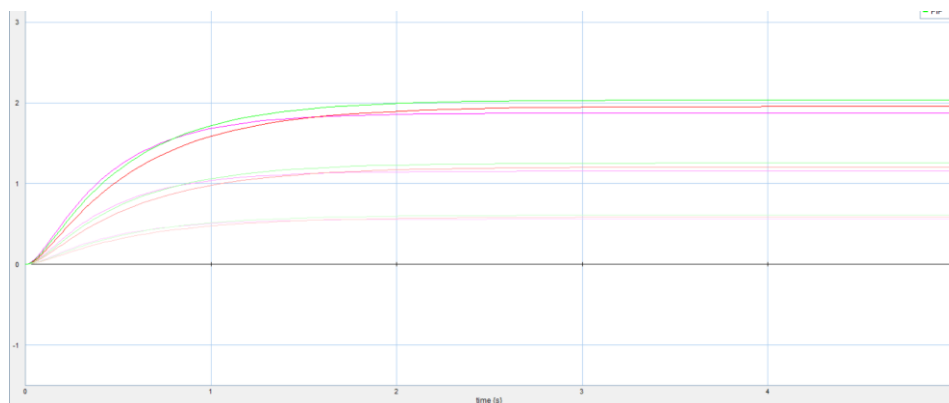
7.4 Thumb Parameters

Parameter	Value	Description
tendon_k	0.0005	Thumb tendon compliance/stiffness parameter
tendon_b	0.05	Thumb tendon damping
Pulley r	1	Thumb pulley transformer ratio
TF r	166	Thumb tendon-to-axis transformer factor
J_thumb_motor_axis	0.00019	Thumb motor-axis inertia
B_thumb_motor_axis	0.027	Thumb motor-axis damping
J_thumb_servo_axis	0.00026	Thumb servo-axis inertia
B_thumb_servo_axis	0.05	Thumb servo-axis damping
C_DIP1	14	Compliance element in thumb motor-axis branch
Servo_limiter c	0.04	Compliance used to limit/resist the servo branch

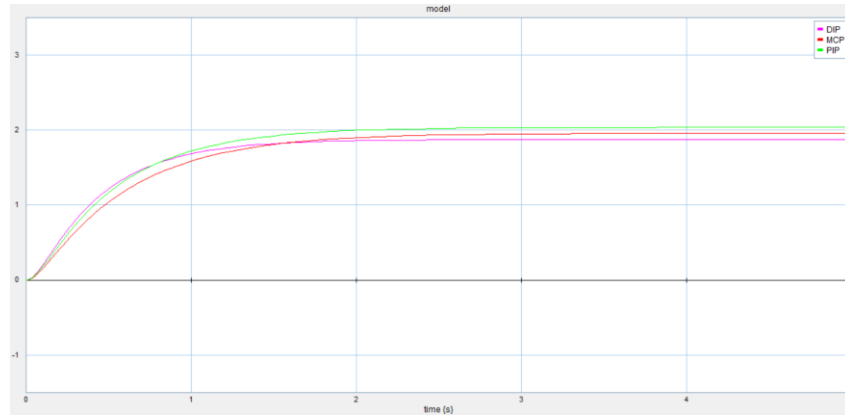
8. Graphical Plots

8.1 Non-Thumb Finger

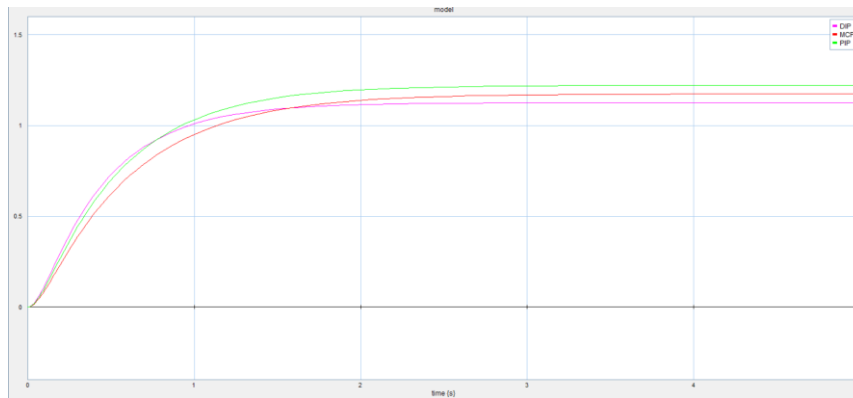
The following plots represent the index, middle, ring, and pinky fingers because all four fingers were modelled using the same parameters. The 12 V input produces the highest final flexion and reaches a gripping angle above 90 degrees. The 7.2 V input gives a lower final angle, while the 3.7 V input gives weak flexion and would not be sufficient for strong gripping



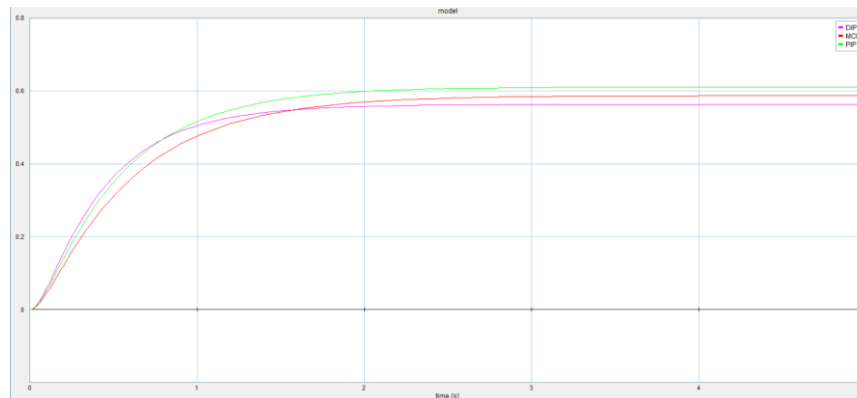
Graph 1. Combined representative finger output angles for the different input voltage cases.



Graph 2. Representative finger MCP, PIP, and DIP angles at 12 V input.



Graph 3. Representative finger MCP, PIP, and DIP angles at 7.2 V input.

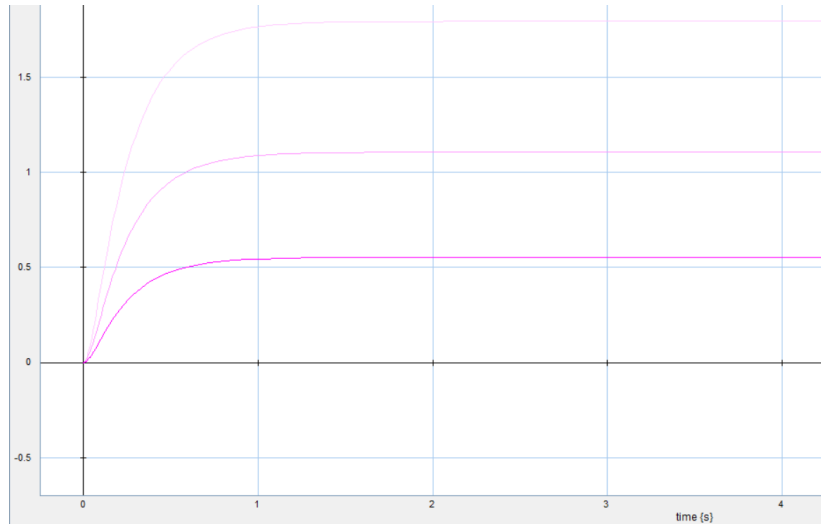


Graph 4. Representative finger MCP, PIP, and DIP angles at 3.7 V input.

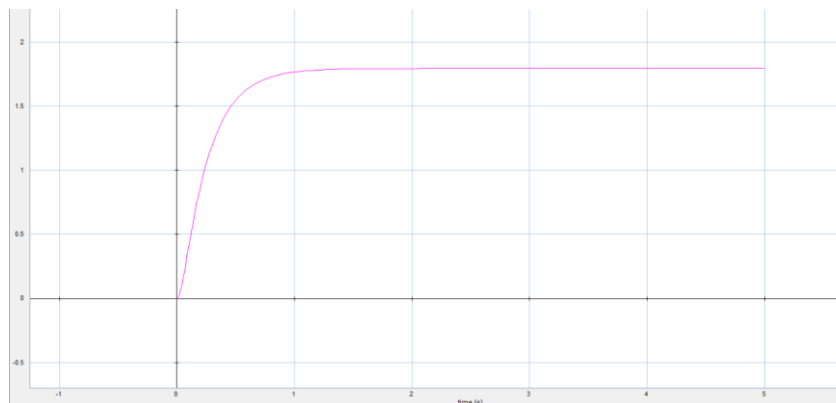
The results show the expected relationship between input voltage and output flexion. Higher voltage increases the available motor effort, which increases tendon force and joint torque. Therefore, the final joint angles become larger. At 12 V the finger flexes fast and reaches the useful gripping range. At 7.2 V the motion is still smooth but the final flexion is smaller. At 3.7 V the hand would have poor gripping ability because the final joint angles are much lower.

8.2 Thumb

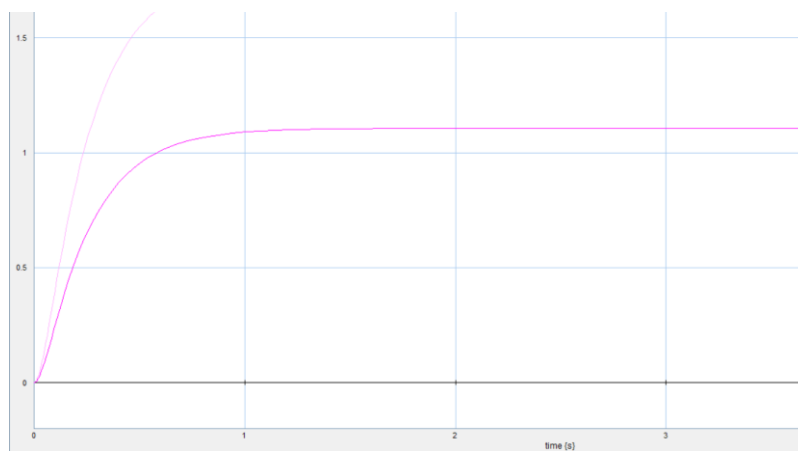
The thumb motor-axis response behaves similarly to the finger flexion response. The 12 V input gives the largest thumb motor-axis angle, while 7.2 V and 3.7 V give smaller final angles. This confirms that battery voltage directly affects thumb gripping as well.



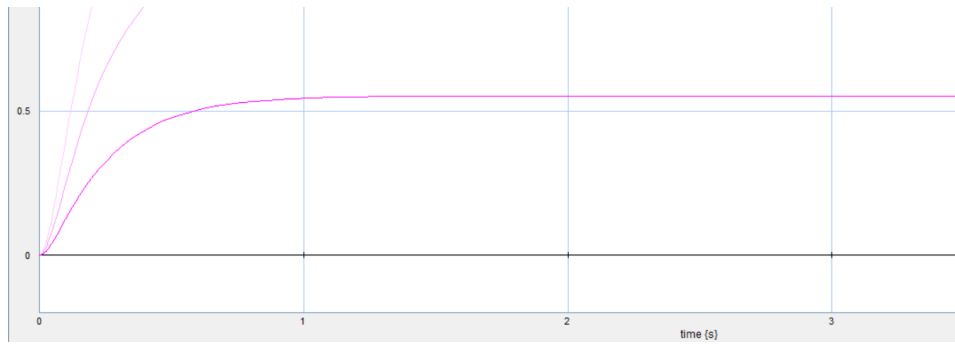
Graph 5. Combined thumb motor-axis angle response for the different input voltage cases.



Graph 6. Thumb motor-axis angle at 12 V input.



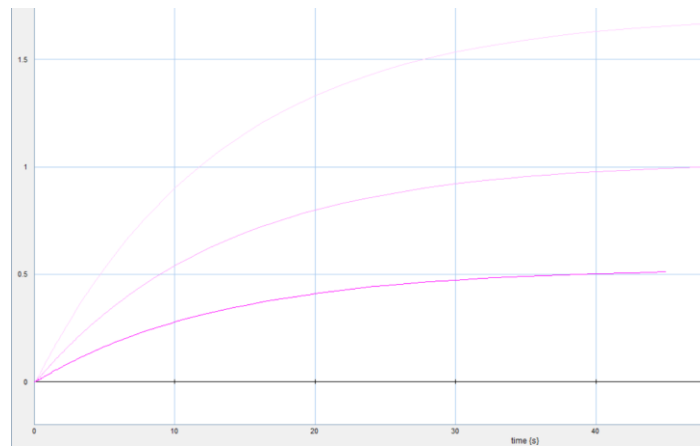
Graph 7. Thumb motor-axis angle at 7.2 V input.



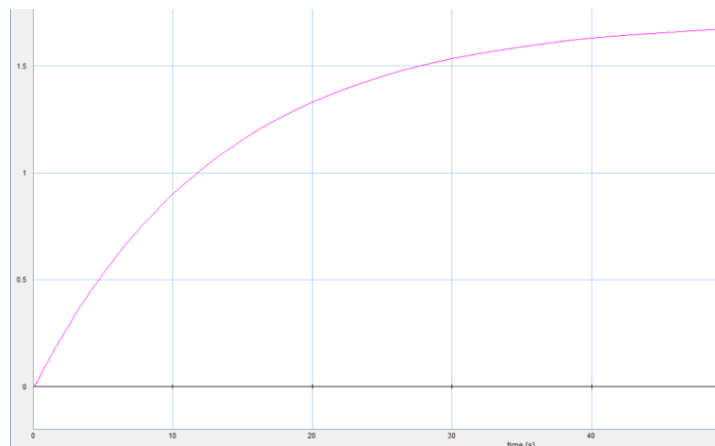
Graph 8. Thumb motor-axis angle at 3.7 V input.

8.3 Thumb Servo Second-Angle Results

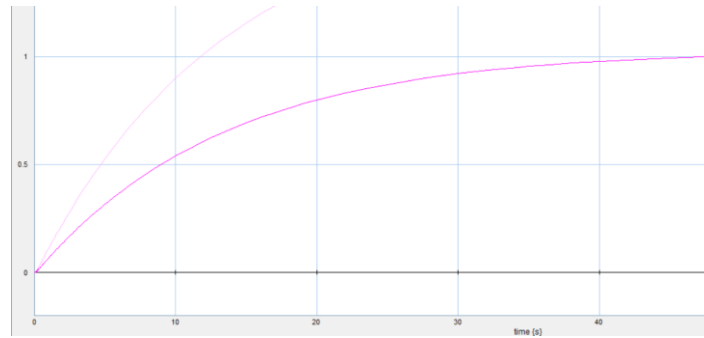
The thumb servo second-angle branch is the main issue in the updated simulation. The output increases very slowly and requires around 40 s to approach its final value, which is not realistic for a prosthetic gripping motion. Changing the passive inertia, damping, compliance, and transformer parameters did not solve this issue. This indicates that the simplified servo model is missing important servo behavior, most probably this is because of the closed-loop position control and realistic servo torque/speed limits not being modelled.



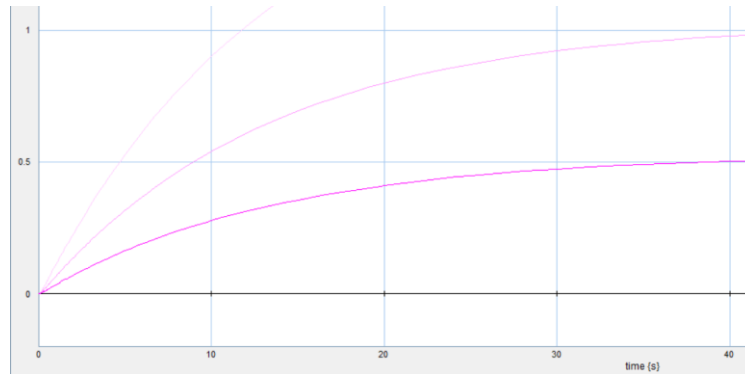
Graph 9. Combined thumb servo second-angle response for the different input voltage cases.



Graph 10. Thumb servo second-angle response at 12 V input.



Graph 11. Thumb servo second-angle response at 7.2 V input.



Graph 12. Thumb servo second-angle response at 3.7 V input.

8.4 Simulation Results Summary

Output	12 V case	7.2 V case	3.7 V case	Main observation
Representative finger	Fast response; reaches useful gripping range above 90 degrees	Moderate flexion; lower final angle	Weak flexion; low final angle	Finger grip strength decreases as battery voltage decreases
Thumb motor axis	Largest final thumb flexion	Reduced final thumb flexion	Smallest final thumb flexion	Thumb motor-axis behavior follows the same voltage trend
Thumb servo second axis	Large final angle but very slow response	Lower final angle and still very slow	Smallest final angle and still very slow	Second thumb angle is not physically realistic in the current simplified model

9. Validation

The validation reference was taken from the team bachelor thesis experiment. In the real prosthetic hand, the hand flexed in approximately 0.835 s when a force of 40 N was applied by the motor. This response was used as the target for the model. The simulated fingers and thumb motor-axis responses at high input voltage reach this flexion range in approximately the same time, but the servo motor axis was very slow in comparison.

The model response was checked against the expected flexion time, steady-state flexion level, and voltage sensitivity. The main tendon-driven finger model is acceptable for studying how input voltage affects flexion. However, the thumb servo second-angle branch is not validated because it does not match the real hand timing.

Validation item	Experimental/expected behavior	Simulation behavior	Conclusion
Finger flexion time	Real hand flexes in 0.835 s under about 40 N motor-applied force	Main finger and thumb motor-axis responses reach the useful flexion range quickly with high voltage input	Acceptable for main flexion study
Battery voltage effect	Lower available voltage should reduce motor effort and grip quality	3.7 V and 7.2 V produce smaller final angles than 12 V	Expected trend confirmed
Thumb second angle	A real servo should move within a short actuation time	Simulated second angle takes about 40 s	Not validated; requires improved servo model

10. References

- Chungsangsatiporn, W., Sithiwichankit, C., Chancharoen, R., Chaichaowarat, R., Ajavakom, N., & Phanomchoeng, G. (2025). *A body-powered underactuated prosthetic finger driven by MCP joint motion*. *Robotics*, 14(8), 107.
- Abbasi, S. H., Mahmood, A., & Saeed, M. T. (2022, July). *Bond graph modeling, simulation and optimal control of a prosthetic finger*. In *8th International Mardin Artuklu Scientific Researches Conference* (pp. 238–244).
- Saeed, M. T., Khaydarov, S., Ashagre, B. L., & Zafar, M. S. (2019). *Comprehensive bond graph modeling and optimal control of an anthropomorphic mechatronic prosthetic hand*. In *2019 IEEE International Conference on Mechatronics and Automation (ICMA)* (pp. 2006–2011).